WIEM RHOUMA

Data Science Engineering Student | Data Engineering, BI and AI/ML
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PROFILE

Data Science engineering student at the Faculty of Sciences of Sfax (FSS) with end-to-end experience across the data lifecycle: data warehousing on Microsoft Fabric (medallion architecture, PySpark, dbt), BI (Power BI, DAX, Row-Level Security), forecasting (Prophet) and applied AI (Text-to-SQL with LLMs and RAG, CNN image classification, clinical prediction on imbalanced data). Also hands-on with Hadoop/Spark, Kubernetes and CI/CD. Seeking a 4 to 6 month Data Engineer / Data Scientist Intern position (PFE, end-of-studies internship) from January/February 2027, in Tunisia.

EDUCATION

Engineering Degree in Data Science, Faculty of Sciences of Sfax (FSS), Tunisia — 2024 – Expected 2027
Research Master's in Computer Science (M2, in parallel), Faculty of Sciences of Sfax (FSS), Tunisia — 2026 – 2027
Integrated Preparatory Cycle in Computer Science, FSS, Tunisia — 2022 – 2024
Scientific Baccalaureate (with Honors), Rafaha High School — 2022

PROFESSIONAL EXPERIENCE

Data Engineering Intern — Jul 2026 – Aug 2026
Biware Consulting, Tunis, Tunisia (hybrid)

- Built a data warehouse on Microsoft Fabric (medallion: Bronze, Silver, Gold) on AdventureWorks 2019: 307,407 rows processed in Silver without loss; profiled 72 tables and designed a constellation schema (2 facts, 6 dimensions)
- Built 11 Silver and 8 Gold models in PySpark and pandas, plus equivalent dbt models (staging, marts, tests) on SQL Server
- Automated the pipeline locally (Python, SQL Server): orchestrator (43.6 s), daily schedule, Power BI refresh via gateway
- Delivered a 5-page Power BI dashboard (DAX, Row-Level Security, 3 roles) and a Prophet revenue forecast (6-fold rolling-origin cross-validation vs a naive baseline; reliable from about 9 quarters)
- Caught data quality issues before reporting (incomplete final quarter; Bikes have no purchase lines, margin not comparable)
- Built a Text-to-SQL assistant (Groq LLM) with 3 safeguards (SELECT-only, single statement, row limit), plus a Flask portal

AI Developer — Feb 2026 – Jun 2026
QueryAgent, Text-to-SQL System | DES Lab, Faculty of Sciences of Sfax, Tunisia

- Extended Vanna AI into a Text-to-SQL system (Groq/Llama-3.3-70B, FastAPI, ChromaDB, PostgreSQL) with question classification, adaptive RAG, EXPLAIN-based SQL validation with automatic retry and JWT authentication
- Cut average token use from about 5,000 to 1,800 per request (-65%) versus base Vanna AI; scored 22/30 (73.3%) against 18/30 (60%) for the raw LLM on an initial 30-question benchmark

Data and Monitoring Systems Intern — Jun 2025 – Jul 2025
General Directorate of Water Resources (DGRE), Tunis, Tunisia | Groundwater Monitoring Platform

- Built a real-time groundwater monitoring platform covering 24 governorates (Angular 17, Firestore, Leaflet, Chart.js): live map, threshold alerts, history, role-based access; deployed on DGRE's infrastructure
- Worked with DGRE's non-IT engineers to gather requirements and validate the platform through user testing
- Cleaned and integrated 4 types of sources (CSV, Excel, PDF, legacy databases) into Firestore; built Python dashboards (Pandas, Matplotlib) with exports in 6 formats

PROJECTS

Kubernetes DevOps Pipeline — 2026
Automated a Kubernetes v1.28 cluster (1 master, 2 workers, 4 VMs) and a CI/CD chain deploying a 3-tier app. *Stack: Vagrant, Ansible, kubeadm, Jenkins, Docker, Gitea, Nexus, Helm, NGINX Ingress, NFS, Linux*

Hadoop and Spark Big Data Pipeline — 2026
3-node high-availability cluster with automatic failover validated by stopping the active NameNode; Spark jobs and MapReduce pipeline. *Stack: HDFS, YARN, ZooKeeper, PySpark, Scala, Apache Spark, MapReduce*

Big Data Analysis with Apache Hive — 2026
Analyzed 1M-row public datasets with Apache Hive: built ORC tables, cleaned the data and cut query I/O by about 110x with date partitioning. *Stack: Hive, HiveQL, ORC, Linux*

SepsisGuard: Early Sepsis Prediction — 2026
Compared 4 models on 1.55M clinical records (1.8% sepsis) with undersampling, lag/rolling features, a patient-level train/test split and GridSearchCV; best (HistGradientBoosting): ROC-AUC 0.80, recall 0.70 on a class-balanced test set. *Stack: Python, XGBoost, Scikit-learn, Data Analysis, Feature Engineering, Machine Learning*

Smart Clinic: Skin Disease Classification (team project, my module) — 2026
Fine-tuned EfficientNetV2-S on DermNet (23 classes, about 19,500 images) with weighted sampler, class weights and label smoothing: accuracy 75.2%, balanced accuracy 67.8%, ROC-AUC 0.92. *Stack: Python, PyTorch*

Population Genetics Analysis, DNA/SNP — 2026
995 individuals from 7 populations: PCA,K-means suggests 5 clusters (silhouette 0.35), SNP-subset robustness test, FST, IBS and π metrics. *Stack: Python, PCA, K-Means, Unsupervised Machine Learning*

YouTube Analytics Data Warehouse — 2025
Collected and built our own channel dataset in Excel (channel type, subscribers) and merged it with 3+ heterogeneous sources into a star-schema warehouse (1 fact, 4 dimensions); ETL implemented twice, OLAP cube and dashboards. *Stack: Talend, MySQL, Python, SQL Server, SSAS, Power BI, PostgreSQL*

NoSQL Pipeline on Wind Turbine SCADA Data — 2025
Kaggle SCADA data to Cassandra (3 tables, 3 materialized views) and Elasticsearch, with Kibana dashboards and anomaly detection. *Stack: Python, Cassandra, Elasticsearch, Kibana*

Face Mask Detection — 2025
Compared 3 CNN architectures (SimpleCNN, DeepCNN, MobileNetV2) on about 9,000 images: 93.6% to 94.7% accuracy on a 3,060-image test set. *Stack: Python, TensorFlow/Keras, CNN*

SKILLS

Data Engineering: Microsoft Fabric (Lakehouse), medallion architecture, ETL/ELT, dbt, Apache Spark (PySpark, Scala), Hadoop, Talend, dimensional modeling (star, constellation), data warehouse, OLAP (SSAS), Hive (HiveQL, ORC, partitioning)
Databases: SQL Server, PostgreSQL, MySQL, Cassandra, Elasticsearch, Firebase Firestore
BI and Analytics: Power BI (DAX, Row-Level Security), Kibana, Pandas, Matplotlib, Excel (Microsoft certified)
ML and AI: Scikit-learn, XGBoost, PyTorch, TensorFlow/Keras, CNN and transfer learning, Prophet, PCA, K-means, LLMs, RAG, ChromaDB, Text-to-SQL, imbalanced data
Cloud and DevOps: Azure, OCI, Docker, Kubernetes, Helm, Ansible, Jenkins, CI/CD, Git, Linux
Programming and Backend: Python, SQL, Scala, FastAPI, C, C++, Spring Boot, Java, Flask, Angular, REST APIs
Soft Skills: Teamwork, problem solving, communication, rigor, autonomy, resilience under pressure

CERTIFICATIONS, ACTIVITIES AND LANGUAGES

Certifications: OCI Data Science Professional (Oracle, 2025); Excel 2019 Associate (Microsoft); APIE (APII, 2025)
Activities: Organizing committee, Cubix 1.0 (2026), CPC, FSS; member, Security Net Club, FSS (2023)
Languages: Arabic (native) | English (fluent) | French (good)